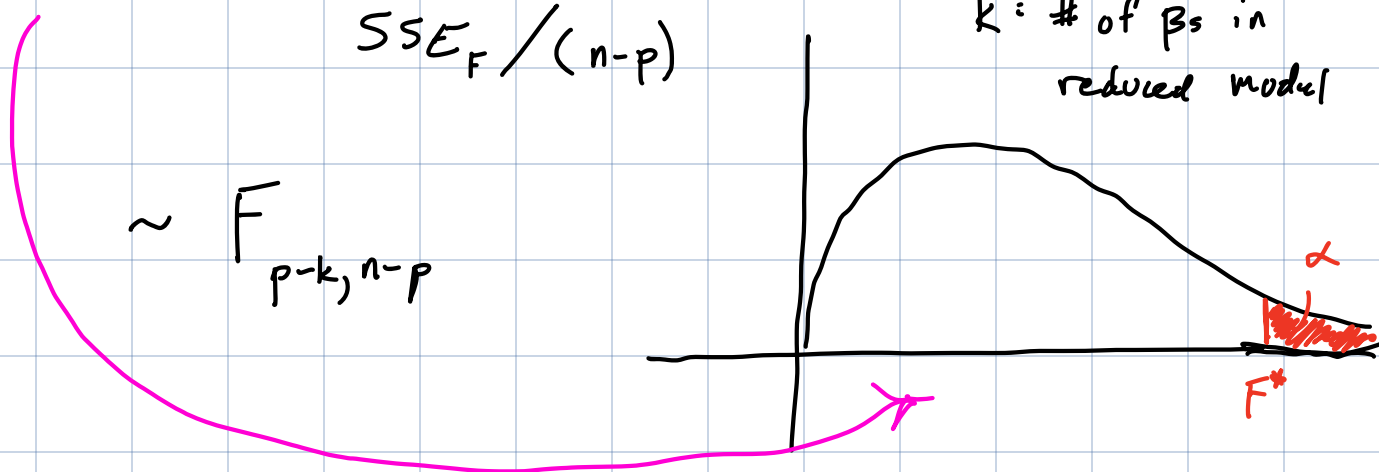


I Partial F-test

$$F = \frac{(SSE_R - SSE_F)/(p-k)}{SSE_F/(n-p)}$$

p : # of β s in Full model

k : # of β s in reduced model



If $F > F^*$ reject H_0 , at least one predictor left out of reduced model is a significant predictor of Y , conditional on the other predictors being in the model.

Specific Predictor Tests

$$H_0: \beta_i = 0$$

$$H_1: \beta_i \neq 0$$

$$F = \frac{(SSE_R - SSE_F)/1}{SSE_F/(n-p)} \sim F_{1, n-p}$$

Related to t -statistic for β_i

$$t = \frac{\hat{\beta}_i - 0}{\widehat{SE}(\hat{\beta}_i)}$$

$$\text{where } \widehat{SE}(\hat{\beta}_i) = \sqrt{\widehat{\text{Var}}(\hat{\beta}_i)}$$

$$= \sqrt{\hat{\sigma}^2 \left[(X^T X)^{-1} \right]_{i,i}}$$

$$\begin{pmatrix} \text{Var}(\hat{\beta}_1) \\ \vdots \\ \text{Var}(\hat{\beta}_{p+1}) \end{pmatrix}$$

Note: $t^2 = F$

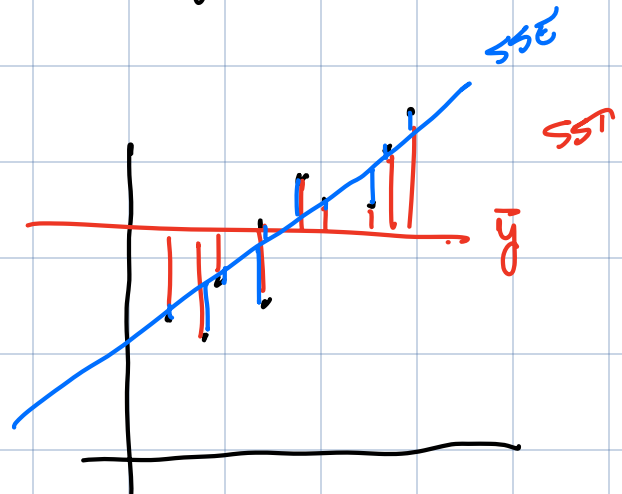
Decision: if $|t| > t_{n-1}^*$ reject H_0 ,

X_i is a significant predictor of y ,

conditional on other predictors being in the model.

II. Adjusted R^2

$$\text{Recall } R^2 = 1 - \frac{SSE}{SST}$$



R^2 increases with additional predictors.

$$R^2_{adj} = 1 - \frac{SSE/(n-p)}{SST/(n-1)} \quad \leftarrow \text{what if } p=1 \text{ (intercept) only}$$

As $p \uparrow$ $SSE/(n-p) \uparrow$ $\frac{SSE/(n-p)}{SST/(n-1)} \uparrow$

$R^2_{adj} \downarrow$
(predictors)

III

Feature Engineering

Transformations

$$y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

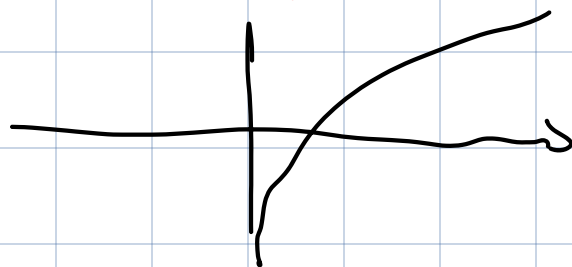
linear in the parameters, but over X 's
can be nonlinear.

① $y_i = \beta_0 + \beta_1 \log(X_i) + \varepsilon_i$

$\log(y_i) = \beta_0 + \beta_1 X_i + \varepsilon_i \quad \leftarrow \beta_1 \approx \% \text{ change in } y \text{ per 1-unit change in } X$

$\log(y_i) = \beta_0 + \beta_1 \log(X_i) + \varepsilon_i$

CAUTION: Be careful if X or $y \leq 0$



(2) Polynomials

$$y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \varepsilon_i$$

↑
not the same
interpretation

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_2 + \beta_4 \log(X_3) + \dots$$

CAUTION: X and X^2 are correlated.

Center X first by taking $X - \bar{X}$

(3) Other transforms

Box-Cox transformation. Try to make y or X more Normal.

$$\text{Transform } y \text{ to } \begin{cases} y^\lambda & \text{if } \lambda \neq 0 \\ \ln y & \text{if } \lambda = 0 \end{cases}$$

IV Categorical Predictors

Insurance data price $\sim$ region price = $\beta_0 + \beta_1 \text{region}$
region = { Northeast, Northwest, Southeast, Southwest }

K-level categorical $\Rightarrow$ K-1 dummy variables

$$K=4$$
$$X = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ \vdots & \vdots & \vdots & \vdots \end{pmatrix}$$

one-hot encoding